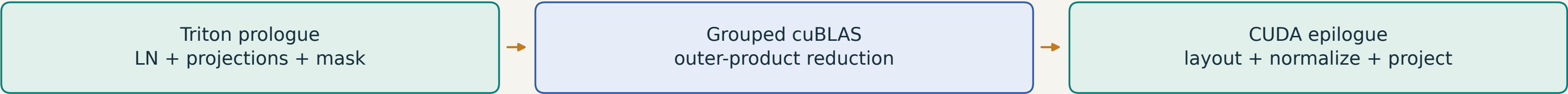


Outer Product Mean: fuse around the sequence reduction

MSA → pair representation · training path [C5]

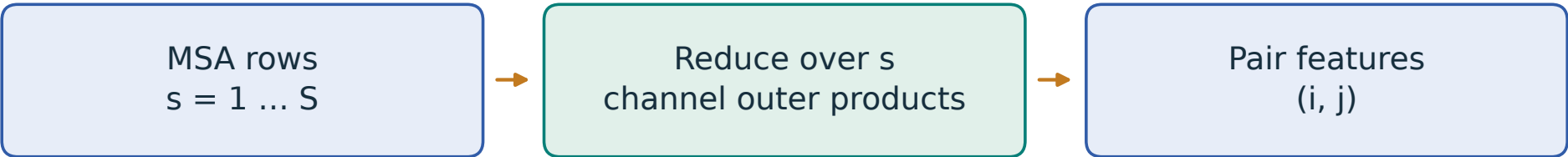
A / EXECUTION AND FUSION



A/B and the product cross global-memory boundaries. Mask-count normalization is computed separately.

B / ALGORITHM AND DATA REUSE

$$O_{ijab} = \sum_s A_{sia} B_{sjb}$$



Backward choice: save O, or recompute O to reduce stored activations.

The performance card uses save O; both variants remain in the data bundle.

HISTORICAL MEASUREMENT

Directly measured full operator F+B

L384

5.858 → 2.028 ms

2.89×

Baseline: Own engine path (job 15640)

Shape: S1024; save O

The sequence reduction is retained as a library call; epilogue fusion combines layout, normalization and projection.